Formulation — planar PoC of impedance-informed human uncertainty, risk-sensitive selection and Ec + BH

Folder: `experiments/human_contact_uncertainty_2d_v1/`. Research simulation only; not a hardware or deployment claim. Numerical values are the frozen defaults in `configs/default.json` (they're reproduced in §13). Every statement is tagged with the guarantee tier from §12:

- **(A)** a proven identity or continuous-time conditional theorem for the stated model;
- **(B)** approximate Gaussian/GP chance-constrained planning (approximation, not a guarantee);
- **(C)** empirical finite-sample frequencies and numerical audits.

What comes from the papers and what is our adaptation is mapped in `literature_mapping.md`. The combination "latent-intention/impedance human prediction + uncertain robot contact coordinate + correlated energy prediction + an independent energy/viability filter" is **our proposed combination**. None of the four papers establishes it, and the comparison with them alone does not establish novelty.

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1. Plant: a planar rigid tool where contact location matters

Generalised coordinates `q = (x, y, θ)`, horizontal plane, **no gravity**. The body is a rectangle, `L × H = 0.40 × 0.10 m`, mass `m`, inertia `I` about the centre of mass (COM). `M = diag(m, m, I)` is constant.

Rotation `R(θ) = [[cos θ, −sin θ], [sin θ, cos θ]]`, `Ω = [[0, −1], [1, 0]]`, and `dR/dθ = R Ω`. A body-fixed point `r` (body frame) has

```
p(q, r) = [x, y]ᵀ + R(θ) r,
ṗ       = J(q, r) q̇,         J(q, r) = [ I₂ , R(θ) Ω r ]   (2×3).
```

A world force `F` applied at `p` produces the generalised force `τ = Jᵀ F = [F; ρ × F]`, where `ρ = R r` and `ρ × F = ρ_x F_y − ρ_y F_x` (because `(R Ω r)ᵀ F = (Ω ρ)ᵀ F`).

- **Task contact (tool tip):** a fixed body point `r_T = (0.12, −0.10)`. It is off-centre, so the wall reaction produces a moment.
- **Human contact patch:** the top edge `r_H(s) = (s, h_e)`, `h_e = +0.05`, with patch coordinate `s ∈ [−0.18, 0.18]` m. `s` is a **coordinate on the robot body**. It is not the human's intended trajectory.

Because `r_H` is affine in `s`, the contact point and its Jacobian are affine in `s` at fixed `q`:

```
p_Hc(q, s) = c(q) + s u(θ),   c(q) = [x, y]ᵀ + h_e R(θ) e₂,   u(θ) = R(θ) e₁,
v_H(q, q̇, s) = ċ + s u̇,       ċ = [ẋ, ẏ]ᵀ − θ̇ h_e R e₁,       u̇ = θ̇ R e₂,
∂J_Hᵀ/∂s · f = [0, 0, (R e₂)ᵀ f]ᵀ .
```

So s changes the human generalised torque $\tau_H = J_H^\top f_H$ only through the moment component. It also changes the robot-side velocity v_H whenever $\dot{\theta} \neq 0$, and so it changes human work. For a force along the edge ($(R\, e_2)^\top f = 0$) the moment does not depend on s. Normal and oblique human forces make s matter. The scenarios therefore include tangential, normal and oblique interaction at several moment arms.

Dynamics (all forces in world frame, actuation at the COM as generalised force u_{act}):

$$M\,\ddot{q} = u_{act} + J_T^\top F_T + J_H^\top f_H - B_R\,\dot{q}\ ,$$

$B_R = \mathrm{diag}(b_x, b_y, b_\theta) \succeq 0$ is the **physical** robot dissipation (bearing/support losses), $D_R = \dot{q}^\top B_R\,\dot{q} \geq 0$. Controller damping is part of u_{act} and therefore of P_a. It is never counted again in D_R. Actuation is bounded componentwise, $|u_{act,i}| \leq u_{max,i}$. u_{act} (called u_{actual} in the task text) is the command **after** every saturation and filter, i.e. exactly what is applied. Zero-order hold at the servo rate (1 kHz).

2. Constitutively passive unilateral task contact

A fixed compliant wall occupies $y < 0$ with outward normal $n = e_y$. Let $p_T = p(q, r_T)$, $v_T = J_T\,\dot{q}$, $\mathrm{gap} = p_{T,y}$, $v_n = v_{T,y}$, $v_t = v_{T,x}$.

$$\begin{aligned}
&\delta = \max(0, -\mathrm{gap})\ , \quad \dot{\delta} = -v_n \quad \text{(used only when } \delta > 0),\\
&\chi(\delta) = \delta\,/\,(\delta + \delta_s)\ , \quad \chi(0) = 0\ , \quad 0 \leq \chi < 1,\\
&f_n = k_n\,\delta + b_n\,\chi(\delta)\,[\dot{\delta}]_+ \qquad \text{(normal force ON the tool, along } +n),\\
&f_t = -\mu\, f_n \tanh(v_t / v_\varepsilon) \qquad \text{(tangential force ON the tool)},\\
&F_T = f_t\, e_x + f_n\, e_y\ ,\\
&S_T = \tfrac{1}{2}\,k_n\,\delta^2\ ,\\
&D_T = b_n\,\chi(\delta)\,[\dot{\delta}]_+^2 + \mu\, f_n\, v_t \tanh(v_t / v_\varepsilon)\ .
\end{aligned}$$

Identity (A). The power flowing from tool into the environment is $p_T = -F_T \cdot v_T = -f_n\, v_n - f_t\, v_t$. In contact ($\delta > 0$), $-f_n\, v_n = f_n\,\dot{\delta} = k_n\,\delta\,\dot{\delta} + b_n\,\chi\,[\dot{\delta}]_+\,\dot{\delta}$, and $[\dot{\delta}]_+\,\dot{\delta} = [\dot{\delta}]_+^2$. Also $-f_t\, v_t = \mu\, f_n\, v_t \tanh(v_t/v_\varepsilon)$. Hence

$$p_T = \dot{S}_T + D_T\ , \quad \text{with} \quad D_T \geq 0 \quad (\text{since } f_n \geq 0 \text{ and } v \tanh(v/v_\varepsilon) \geq 0).$$

Unloading ($\dot{\delta} < 0$): the damping term is switched off, $f_n = k_n\,\delta \geq 0$, and the identity holds with only frictional dissipation. Nothing is credited for elastic release. *Separation* ($\delta = 0$): $f_n = f_t = S_T = D_T = p_T = 0$. *Transitions:* $S_T = \tfrac{1}{2}\,k_n \max(0, -\mathrm{gap})^2$ is C^1 in gap, and $f_n \to 0$ as $\delta \to 0$ because $\chi(0) = 0$. Forces stay bounded, there are no impulses, and the identity holds almost everywhere. The force is never adhesive ($f_n \geq 0$). The damping term needs no assumption that clipped Kelvin–Voigt forces dissipate $b_n\,\dot{\delta}^2$. The dissipation actually produced is $b_n\,\chi\,[\dot{\delta}]_+^2$. Contact parameters are fixed and known in the main experiments. A separate stress test (§10.6) uses a known friction interval and a conservative lower credit.

Only the tip interacts with the wall. The auditor checks that no other body corner penetrates the wall (domain check, reported).

3. Composite storage and power balance (A)

Robot kinetic storage $S_R = \tfrac{1}{2}\,\dot{q}^\top M\,\dot{q}$ (no gravity), task storage S_T, composite $S = S_R + S_T \geq 0$. With $P_a = u_{act}^\top \dot{q}$, $p_H = -f_H^\top v_H = -\tau_H^\top \dot{q}$, $D_R = \dot{q}^\top B_R\,\dot{q}$:

$$\dot{S}_R = P_a - p_T - p_H - D_R\,, \qquad \dot{S}_T = p_T - D_T\,,$$
$$\dot{S} = P_a - p_H - D_R - D_T\,.$$

$p_H > 0$ means robot–to–human output. f_H is the force exerted **by the human on the robot** at the robot material point $p_{Hc}(q, s)$. The human power uses the material–point velocity $v_H = J_H(q, s)\,\dot{q}$. If a location estimate migrates, its migration velocity $\dot{s}$ is **not** added to v_H.

4. Ec + BH accounts (preserved, A)

Account definitions and sign conventions are taken from the existing Ec + BH mainline (workflow `human-port-ecbh-20261029`, `FORMULATION.md`, read–only). They are re–implemented here for a different plant.

$$\dot{E}c = c_T + c_R - P_a\,, \quad 0 \leq c_T \leq D_T\,,\ 0 \leq c_R \leq D_R\,,\ Ec \geq E_{min}\,,$$
$$\dot{B}H = -p_H\,, \quad BH \geq 0$$

(net human–output account; human input replenishes it).

$c_R = 0$ throughout (D_R is physical but not reimbursed). c_T is the **realised** credit from measured state and the known contact law (§4.1). A predicted or mean D_T is never booked as credit. There is no BU account, no task–authorisation coefficient, no residual–energy subtraction, and no implicit reset or time refill of BH. Release does not refill BH.

With $V = S + Ec - E_{min}$:

$$\dot{V} = -p_H - (D_T - c_T) - (D_R - c_R) \leq -p_H\,.$$

If every prefix satisfies $Ec \geq E_{min}$, $BH \geq 0$, and the credit is valid, then $W_{H,net}(t) = \int_0^t p_H \leq \min(V(0), BH(0))$. BH is a constraint ledger and is **not** added to V. GP confidence, likelihood or risk preference never authorises credit above the valid physical bound.

4.1 Discrete booking (what the software does)

Servo period $h_s = 1$ ms, zero–order–hold command u_k, exact encoder sampling of $q, \dot{q}$ (stated assumption, as in the existing mainline).

- **Actuator work:** within a tick u is constant, so $\int P_a\,dt = u_k^\top (q_{k+1} - q_k)$ **exactly**. The software books this value. Booked actuator work is exact under exact encoders.
- **Credit:** $C_k = h_s \cdot \min(D_T(x_k), D_T(x_{k+1}))$ ("min–endpoint" conservative quadrature). This lower–bounds $\int D_T$ over the tick when D_T has no interior dip below both endpoints within 1 ms. That is not proven. The independent auditor checks the prefix condition $\Sigma_{j\leq k} C_j \leq \int_0^{t_{k+1}} D_T$ on the fine integration grid and reports any violation (C).
- **Ec ledger:** $Ec_{k+1} = Ec_k + C_k - u_k^\top(q_{k+1} - q_k)$. It is exactly the account at tick boundaries. Intra–tick minima are audited (C).
- **BH ledger (booked):** $BH^{book}_{k+1} = BH^{book}_k + \tfrac{1}{2}\,h_s\,(y_k^\top \hat{v}_k + y_{k+1}^\top \hat{v}_{k+1})$, $\hat{v} = J_H(q, \hat{s})\,\dot{q}$, y = measured force (§6). This is the **known software ledger**. It is not the true BH.

4.2 True-account estimation error (both accounts)

Ec: under exact encoders, booked actuator work equals true actuator work, so `Ec_true = Ec_book` at tick boundaries. The only Ec-related estimation error is credit validity (previous bullet). It is audited, not assumed. In the parameter-mismatch stress test, the uncertainty about `D_T` is handled by a conservative lower credit (§10.6). The **future** Ec is uncertain through uncertain robot motion and is propagated (§9).

BH: with `y = f + b_f + ε_m` and `v_H(s) − v_H(ŝ) = θ̇ R e₂ (s − ŝ)`, to first order (dropping the second-order product of measurement and location errors):

```
e_BH := BH_true − BH_book = Σ_k h_s [G_k (s − ŝ_k)] − b_fᵀ Δp̂ − n_e ,
G_k = θ̇_k y_kᵀ R_k e₂ ,   Δp̂ = Σ_k h_s v̂_k ,   n_e ~ N(0, Σ_k h_s² v̂_kᵀ Σ_m v̂_k).
```

`e_BH` is affine in the latent `(s, b_f)` plus independent white-noise accumulation `n_e`. It is therefore **correlated with the location/force posterior** of §7. That is how initially independent errors become correlated through inference. The software tracks `G_tot = Σ h_s G_k`, `Γ = Σ h_s G_k ŝ_k`, `Δp̂`, and `var(n_e)` per contact event. Completed events are frozen with their posterior error. Uncertainty about past work is **never** reset to zero at a planning call. (Independence of the 1 kHz ledger noise from the 25 Hz GP samples is an approximation. The shared samples carry 1/40 of the ledger-noise variance.)

5. Backup reserve and the continuous-time reference filter

5.1 Lemma (passive zero-input backup, A)

Horizontal plane, no gravity, `S ≥ 0`, and the identities of §3. Under ideal immediate `u_act ≡ 0` on `[t₀, t]`, `P_a = 0`, so `Ṡ = −p_H − D` with `D = D_R + D_T ≥ 0`. Integrating,

```
∫_{t₀}^{t} p_H dτ = S(t₀) − S(t) − ∫ D ≤ S(t₀)    for every prefix t ≥ t₀,
```

whatever the human does. Hence, with `R_H(x) := S(x)` and `h_H := BH − S`: if `h_H(t₀) ≥ 0`, then after switching to zero input, `BH(t) = BH(t₀) − ∫ p_H ≥ BH(t₀) − S(t₀) ≥ 0` for all future prefixes. Under zero input `Ėc = c_T + c_R ≥ 0`, so `Ec ≥ E_min` is also preserved. `BH ≥ S` is a **sufficient** reserve for this passive backup's energy bound. It is conservative, not the minimum reserve. Moreover,

```
ḣ_H = Ḃ H − Ṡ = −p_H − (P_a − p_H − D) = D − P_a     (p_H cancels exactly).
```

What the lemma does not say. Zero input is **not** a safe physical stop. The tool may keep moving or drift. The human can still be pushed by stored elastic energy released from the contact (bounded by `S(t₀)`). Contact force can make excursions. Stopping at zero velocity does not remove elastic storage. The lemma certifies only the stated energy accounts. It does not bound force, peak power or injury.

5.2 Proposition (continuous-time reference filter, A — conditional)

Assume:

- exact current `Ec`, `BH`, `S` and a lower bound `0 ≤ D_lo ≤ D`;
- `u_act = 0` is admissible (it is in the box), and these are the only hard constraints;

- the closed loop has absolutely continuous solutions.

Choose `u_act(t)` so that, with rates `λ_H, λ_E > 0`,

```
(i)  P_a ≤ D_lo + λ_H h_H ,      (ii) P_a ≤ c_T + c_R + λ_E (Ec − E_min).
```

Then (i) gives `ḣ_H = D − P_a ≥ −λ_H h_H`, and by the comparison lemma `h_H(t) ≥ h_H(0) e^{−λ_H t}`. (ii) gives `Ec(t) − E_min ≥ (Ec(0) − E_min) e^{−λ_E t}`.

Initial–condition requirement: `h_H(0) = BH(0) − S(0) ≥ 0` and `Ec(0) ≥ E_min`. Then `h_H ≥ 0`, `BH ≥ S ≥ 0` and `Ec ≥ E_min` for all t. **Feasibility:** `u = 0` satisfies (i)—(ii) whenever the accounts are in the set, so the constraint set is non–empty inside the box. If the initial condition fails (`h_H(0) < 0`, §10 "infeasible initialisation"), (i) forces `P_a ≤ D − λ_H|h_H|`. This is active braking or energy extraction. Whether it is achievable within the box is **not** guaranteed, and such runs are labelled infeasible–initialisation, not as covered by the proposition.

Scope. Adding slew limits, force caps, workspace limits or gravity compensation changes the claim, because `u = 0` may no longer be admissible. The implementation (§11) is sampled–data (ZOH, 1 ms) and uses estimated accounts. It is therefore a **numerical validation (C)** of this continuous–time reference, not an intersample proof. An intersample bound would need `|P_a(t) − P_a(t_k)| ≤ |u_k| ‖q̈‖_∞ h_s` with a bound on `q̈` that includes the human force; this PoC does not establish one. Observed intersample deficits are reported.

6. Human truth model (simulator only; inaccessible to the controller)

Following P1 Eqs. (7)–(8) with diagonal impedance and **no human inertia** (we omit P1's inertia term `M ë`). We model `f_H`, the force exerted **by the human on the robot**:

```
f_H(t) = w(t) · [ K_H (a_H(t) − p_Hc(t)) + D_H (ȧ_H(t) − v_H(t)) + g_true(χ_t) +
ε_phys(t) ] ,
```

- `a_H(t)` is the human's latent intended contact–point trajectory (world frame). It is generated from **intent families** (§10), not sampled from the fitted GP.
- `K_H, D_H` are positive diagonal. They're drawn per episode, lognormal around the nominal values, plus held–out gain–mismatch cases.
- `g_true(χ) = −c_g tanh(χ / v_g)` per axis is a residual nonlinearity in the patch velocity `χ = ċ`. It is switched off in the first validation stage.
- `ε_phys` is physical force variability, an OU process (`σ_p`, `τ_p`), sampled once per episode on a fine grid. It acts on the plant.
- `w(t) ∈ [0, 1]` is the engagement. At grasp, `a_H(t_on) = p_Hc(t_on)` (no pre–load), so the force starts continuously. Release is a 20 ms linear ramp to zero at `t_off`. It is triggered by the scenario script or by `|f_H| > F_rel` (the human lets go).
- The true location `s` is fixed for each contact event and drawn from the scenario.

Positive impedance gains with a moving intention do **not** make the human passive. The human injects energy whenever `a_H` moves. We exclude human inertia. Adding `M_H(ä_H − p̈_Hc)` would couple the robot and human dynamics implicitly and needs a new derivation. No future ground–truth acceleration is fed to the predictor.

7. Structured (impedance-informed) latent-intention GP — our specialisation of P1

7.1 Observation model

These quantities are available to the controller:

- exact robot state;
- the measured human force `y_j = f_H(t_j) + b_f + ε_m,j`, white `ε_m ~ N(0, σ_m² I)` at 1 kHz; this measurement noise is **not** applied to the plant; `b_f` is a persistent sensor bias, zero in nominal runs;
- location readings `s_obs = s + b_s + ε_s` at 25 Hz, where `b_s` is a persistent per-episode error and `ε_s` is white;
- a binary contact-presence flag.

The truth intention `a_H`, the true `s`, gains, the residual, the OU path and future data are inaccessible. Oracle runs are labelled.

7.2 Prior and its linear structure

For one contact event starting at `t_on`, and per axis `i ∈ {x, y}`:

```
a(t) = p_Hc(t_on) + ã(t) = c_on + s u_on + ã(t) ,   ã_i ~ GP(0, k_a) ,
k_a(t,t') = σ_a² exp(−(t−t')²/(2ℓ_a²)) + σ_c² + σ_l² ((t−t_on)(t'−t_on))/T_l² ,
```

We use two more latent groups:

- residual `g_i(χ) = φ(χ)ᵀ w_i` with Gaussian RBF features `φ`, weights `w_i ~ N(μ_w,i, Σ_w,i)` (the frozen offline posterior); this is an explicitly labelled finite-feature approximation;
- physical noise `ε_p,i ~ GP(0, σ_p² e^{−|t−t'|/τ_p})`.

Location, force-bias and location-bias priors are `s ~ N(μ_s0, σ_s0²)`, `b_f ~ N(0, σ_bf² I)`, `b_s ~ N(0, σ_bs²)`. All latent groups are a-priori independent.

For fixed gains `K, D` and the **conditioned, deterministic** (measured) past robot trajectory, `L_t = K + D d/dt` gives

```
f(t) = K(c_on − c(t)) − D ċ(t) + s h(t) + K ã(t) + D ȧ(t) + Φ(χ_t) w + ε_p(t),
h(t) = K(u_on − u(t)) − D u̇(t)                                    (2-vector),
```

All terms are **linear** in `(ã, s, w, b_f, ε_p)`. The residual input `χ_t = ċ(t)` (patch-reference velocity) does not depend on `s`. Prior force mean and covariance per axis:

```
m_f(t)     = K(c_on − c(t)) − D ċ(t) + μ_s0 h(t) + Φ(χ_t) μ_w ,
K_f(t,t')  = K² k_a + K D (∂₁k_a + ∂₂k_a) + D² ∂₁∂₂k_a + k_p + Φ_t Σ_w Φ_{t'}ᵀ
             + σ_s0² h(t) h(t')ᵀ (couples axes) .
```

This equals the task text's `L_t K_a L_t'ᵀ + K_g` (here `K_g = Φ Σ_w Φᵀ`) plus the OU and location terms. It is P1 Eq. (17)–(18) with the inertia set to zero and deterministic gains (`σ_d = σ_k = σ_m = 0`). For the stationary SE part the `K D` cross-term cancels on the diagonal. The observation adds `σ_bf² +`

$\sigma_{m^2}\,\delta_{jl}$. The location summary $\bar{s}$ (mean of n_s readings) has $\mathrm{Var}(\bar{s}) = \sigma_{s0^2} + \sigma_{bs^2} + \sigma_{\varepsilon s^2}/n_s$ and $\mathrm{Cov}(y_j, \bar{s}) = \sigma_{s0^2}\, h(t_j)$.

Intention–force cross-covariance (P1 Eq. (25) with $\mu_m = 0$; ∂_1 = derivative in the first argument):

$$\begin{aligned}
&\mathrm{Cov}(y_{j,i}, \tilde{a}_i(t*)) = K_i\, k_a(t_j,t*) + D_i\, \partial_1 k_a(t_j,t*),\\
&\mathrm{Cov}(y_{j,i}, \mathring{a}_i(t*)) = K_i\, \partial_2 k_a(t_j,t*) + D_i\, \partial_1\partial_2 k_a(t_j,t*),\\
&\mathrm{Cov}(y_j, s) = \sigma_{s0^2}\, h(t_j),\quad \mathrm{Cov}(y_{j,i}, w_i) = \Phi_j\, \Sigma_{w,i},\quad \mathrm{Cov}(y_{j,i}, b_{f,i}) = \sigma_{bf^2},\\
&\mathrm{Cov}(y_{j,i}, \varepsilon_{p,i}(t_{now})) = k_p(t_j, t_{now}).
\end{aligned}$$

Units: k_a [m^2], ∂k_a [m^2/s], $\partial_1\partial_2 k_a$ [m^2/s^2]. Every force–covariance block is [N^2]. This is tested, together with finite-difference derivatives.

7.3 Joint conditioning (exact Gaussian under the stated conditions)

The targets are

$$\zeta = [\tilde{a}_x, \mathring{a}_x, \tilde{a}_y, \mathring{a}_y \text{ at the } N \text{ horizon nodes}; s; w_x; w_y; b_f; \varepsilon_p(t_{now})]\ .$$

We stack the window observations $Y = [y\ (2n); \bar{s}]$ and apply Gaussian conditioning (P1 Eq. (26), with the minus sign and transpose verified):

$$E[\zeta \mid Y] = \mu_\zeta + C^\top K_{YY^{-1}}\,(Y - m_Y), \qquad \mathrm{Cov}[\zeta \mid Y] = \Sigma_\zeta - C^\top K_{YY^{-1}}\, C\ .$$

This is **one** joint conditioning. After observing common force data, the intention, residual weights, location, bias and OU state become correlated. We keep all of those cross-covariances. We never condition two GPs separately and add their variances. The posterior is exact Gaussian because:

- the gains are fixed;
- the past robot trajectory is conditioned (known);
- χ is deterministic;
- p_{Hc} is affine in s.

Approximations, stated. Integrating over uncertain gains, uncertain future robot motion (feedback), or s inside nonlinear functions (e.g. future rotations) destroys exact Gaussianity. Future propagation (§9) is therefore approximate (B): it is either Monte Carlo over consistent joint samples or first-order linearisation. The Gaussian prior on s is a local approximation on a bounded patch. Samples are **not** clamped. The predicted probability mass outside the patch is logged; when it exceeds 1% the run is flagged.

7.4 Identifiability and confounding

- s and a constant intention offset are confounded. $K(\tilde{a} - s\,u)$ is invariant to $(\tilde{a}, s) \to (\tilde{a} + \Delta u, s + \Delta)$ when u is constant. Force data inform s only through **rotation** ($u(t) \neq u_{on}$, $\dot{u} \neq 0$). The independent location sensor is therefore essential. Its persistent bias b_s sets a floor on location uncertainty that does not average out.
- A constant force bias b_f is confounded with an intention offset: $K\,\Delta\tilde{a} = b_f$.
- A flexible residual $\Phi\, w$ and the intention $\tilde{a}$ can explain the same force. The residual is pinned by **cross-episode** training data, because its input is the patch velocity rather than time. Within one episode the decomposition is not unique. We report the posterior correlation between w and $\tilde{a}$. We do not claim unique recovery of both.

- Gains, intention and location are **not** jointly inferred online from a single force observation. Gains are fixed: nominal values are chosen offline on training trajectories with exogenous robot excitation (marginal likelihood). Joint gain inference is deferred.

7.5 Baselines (matched data/window)

- **Naive force GP** (P1 Alg. 1 step 1 analogue): `f_i ~ GP(0, k_SE([t, χ])) + white`, hyperparameters by maximum likelihood on the same training split. It responds to candidate motion only through learned correlation with `χ`.
- **Gauss–Markov (C2):** per axis, `f = μ_f + η`, η an OU (`τ_f, σ_f` fitted offline), `μ_f` = window mean with its standard error. It is an exogenous force forecast and does **not** respond to candidate robot motion.
- **Deterministic (C1):** hold the last (window-averaged) measured force; location `ŝ`; no covariance.

8. Generalised human torque uncertainty (illustration, first order)

At known `q`, with `[s, f]` jointly uncertain:

$$\delta\tau_H \approx G_s\,\delta s + J_H^{\top}\,\delta f\,, \quad G_s = \partial J_H^{\top}/\partial s \cdot \bar{f} = [0,\,0,\,(R\,e_2)^{\top}\,\bar{f}]^{\top}\,,$$
$$\mathrm{Cov}(\tau_H) \approx [G_s,\,J_H^{\top}]\,\mathrm{Cov}([s;\,f])\,[G_s,\,J_H^{\top}]^{\top}\,.$$

This is verified against Monte Carlo (tests and figure `moment_arm`).

9. Belief propagation over a short horizon (B)

Candidate policies are closed-loop feedback laws (§10.1), so the predicted human force depends on the predicted robot motion. The predictor evaluates `f_H(x, s, ζ)` **inside** the rollout. It is never an exogenous force forecast after conditioning at the current state.

Random inputs. These are drawn jointly:

- the GP posterior vector `ζ` from §7.3, which includes `s`, `b_f`, `w` and the current OU state, with all cross-covariances;
- OU innovations per stage, `ξ_k ~ N(0, I)`;
- the past-work error `e_BH` from §4.2, affine in the same `s`, `b_f`, plus `n_e`.

The augmented random vector is `η = [ζ; ξ; n_e]`. The stage force uses `a(t_k + τ) ≈ a_k + τ ȧ_k` within a stage. This is first-order in $\tau \leqslant 20$ ms, and both `a_k` and `ȧ_k` are jointly sampled.

State. `z = [q, q̇, Ec, BH_true, W_H, controller states (x_ref, ∫ force error), J]`. The latent function values are static, so the augmented state of the task text, `[x, s_H, human latents, Ec, BH, ctrl]`, is represented as `z` plus `η`. Their covariance is carried through the sensitivity `Φ_k = ∂z_k/∂η`.

Monte Carlo (nonlinear check / objective). `N_s` joint samples of `η` (common random numbers across candidates):

- **one** posterior function sample per rollout;
- one location per rollout;
- OU innovations evolve per stage.

Each sample is rolled out through the deterministic closed-loop step `z_{k+1} = G_Δ(z_k, η)` (semi-implicit Euler, 5 ms substeps, known contact law, saturation). GP epistemic error is **not** redrawn per

step.

First-order propagation. The mean trajectory is $\bar{z}_{k+1} = G_\Delta(\bar{z}_k, \bar{\eta})$, and $\Phi_{k+1} = A_k \Phi_k + B_k E_k$, where:

- $A_k = \partial G/\partial z$ **includes the feedback-law derivatives**;
- $B_k = \partial G/\partial(\text{local } \eta)$;
- E_k selects the components of η used in stage k.

Both A_k and B_k come from central finite differences on the vectorised step. Then $\mathrm{Cov}(z_k, z_l) = \Phi_k \Sigma_\eta \Phi_l^\top$. This retains the full cross-time covariance, and it is equivalent to $P_{k+1} = A\,P\,A^\top + L\,Q\,L^\top$ on the augmented state.

Integrated work. Predicted human work uses the same predicted trajectory: $W_H = \int -f_H^\top J_H \dot{q}$, $\mathrm{Var}(\Sigma_i W_i) = \Sigma_{i,j} \mathrm{Cov}(W_i, W_j)$ (cross-time terms kept), and $\mathrm{Var}(BH_{next}) = \mathrm{Var}(BH) + \mathrm{Var}(W_H) - 2\,\mathrm{Cov}(BH, W_H)$. The past-work error is correlated with s, hence with future work.

Mean-power correction. $E[p_H] = -E[f]^\top E[v] - \mathrm{tr}\,\mathrm{Cov}(f, v)$. First-order mean propagation evaluates $-\bar{f}^\top \bar{v}$ and misses the trace term. We report it from MC. With human damping, $\mathrm{Cov}(f, v) \approx -D\,\mathrm{Var}(v)$, so first order *under*-predicts expected robot-to-human power.

Monte Carlo and the first-order method are compared on mean, covariance and **work tails** (empirical quantiles), not only on force RMSE. Sigma points or MC moments do not certify tail probabilities.

10. Planning: risk-neutral vs risk-sensitive selection with separate constraints (B)

10.1 Decision space: finite candidate feedback policies

The nominal hybrid controller (all controllers) acts at the tool tip:

- **tangential impedance** to a reference x_{ref} advancing at $\sigma_v\, v_{nom}$, with stiffness $\sigma_K K_t$, damping D_t, and friction feed-forward from the measured contact force;
- **normal force PI** to $\sigma_F f_d$, plus normal damping;
- **orientation PD** to $\theta_{ref} = 0$.

The tip force is mapped by $J_T^\top$, the orientation torque is added, and the result is saturated. The normal channel also has an **approach-speed limiter**: extra damping $D_{brake} \cdot [-v_{app} - v_n]_+$, active only when the tip moves toward the wall faster than v_{app}. The force integrator also leaks out of contact. Both were added after a smoke-test failure; see `decision_log.md`.

The candidate set is:

- $\sigma_v \in \{0, 0.5, 1, 1.5\} \times \sigma_K \in \{1, 0.35\} \times \sigma_F \in \{1, 0.5\}$ (16 policies);
- the **passive backup** $u \equiv 0$ (the policy of §5.1);
- a strictly dissipative **brake** $u = -D_{brk}\,\dot{q}$, for which $P_a = -\dot{q}^\top D_{brk}\,\dot{q} \le 0$, so $\dot{h}_H \ge 0$ and $\dot{E}c \ge 0$.

The brake was added because, under the passive backup, the free tool drifted and rotated until re-engaging the task needed more energy than the remaining BH. That is the "zero input is not a safe physical stop" caveat in practice.

The selected policy runs at the 1 kHz servo rate for one 20 ms control period (receding horizon, N = 15 stages). The solver is **finite enumeration** of these 18 candidates. It is not a QP or NLP, and there is no claim of optimality over continuous inputs.

10.2 Objective (separate from safety)

The stage cost `J` is dimensionless, with each term normalised by a stated scale:

- progress shortfall against the **nominal task schedule**, so slowing down costs progress;
- normal-force error;
- orientation error;
- effort `‖u‖²_W`;
- control increments.

`J` is averaged over the horizon. A separately described **energy-margin (task-viability) cost**, `w_res·max(0, (R_task − h_H,N)/R_task)²` at the horizon end, discourages spending the human budget below what continued wiping needs (`R_task = 0.03 J`, `w_res = 2`). This term is a performance term. It is identical in C1–C4, it is not a constraint, and it is never a substitute for one. The safety constraints (§10.3) are separate and are never traded off inside `J`.

- risk-neutral (C3): `ρ₀(J) = E[J]` (MC mean);
- risk-sensitive (C4): `ρ_β(J) = (1/β) log E[exp(β J)]`, evaluated with stable log-sum-exp over MC samples with equal, non-negative weights. The delta-method sampling standard error and the effective sample size of the exponential weights are logged.

Risk mapping: P3/P4 use `γ(θ) = −(2/θ) ln E[exp(−θJ/2)]`. With `β = −θ/2`, `ρ_β = γ(θ)`. So **β > 0 is risk-averse**, and `ρ_β ≈ E[J] + (β/2) Var(J)` for small β (P3 Eq. (11): `γ ≈ E[J] − (θ/4) Var[J]`). This mean—variance expansion is an approximation and is not used as the objective.

MGF domain: the empirical estimator is always finite. An analytic exponential-quadratic evaluation for Gaussian `z` and quadratic `J = zᵀQz` would need `I − 2βΣ^{1/2}QΣ^{1/2} ≻ 0`. We make no analytic claim.

Compliance direction is **measured**, not assumed: we log the selected `σ_K`, `σ_v`, `σ_F`. P4 reports that its risk-averse controller stiffens.

10.3 Chance constraints (approximate, B)

The constraints are evaluated on first-order moments, with an MC-moment ablation. For every candidate and every stage prefix `k = 1..N`:

```
E[Ec_k] − κ(α_E) std(Ec_k) ≥ E_min ,
E[h_H,k] − κ(α_H) std(h_H,k) ≥ 0 ,   h_H,k = BH_true,k − S_k ,
```

- `h_H` is tightened **as a single quantity**, so the covariance between `BH_true` and `S` is kept. We never tighten BH and S separately and subtract.
- `BH_true,k = BH_book + e_BH − W_H(0..k)` includes the uncertain past-work error.
- `κ(α) = Φ⁻¹(1 − α)` with `α_E = α_H = 0.01` per planning call.

This per-call level is **not** a mission guarantee. The mission-level violation frequency over executed episodes is measured (C), with a stated budget. A horizon-level budget reset at each replan is not a mission guarantee.

Selection: among candidates satisfying all constraints, minimise `ρ_β`. If none satisfies them, select the candidate with the largest minimum normalised margin and log `planner_infeasible`. Then the servo filter acts.

10.4 Baselines with the same candidates, horizon, constraints and budgets

ID	predictor	constraints	objective	servo filter
C0	none (nominal policy σ=(1,1,1))	none	—	**none (unprotected reference)**
C1	deterministic hold-force, $\hat{s}$	mean only (κ = 0)	E[J] (1 sample)	ledger, no margin
C2	Gauss–Markov force + Gaussian `s`	tightened (κ)	E[J]	ledger + belief margin
C3	structured latent-intention GP	tightened (κ)	E[J]	ledger + belief margin
C4	as C3	as C3	ρ_β, β > 0	as C3

The diagnostic **ORACLE-BH≥S** run is a separate row. It runs the nominal policy with the reference filter of §5.2 using **true** accounts. It has truth access and is reported separately. All comparisons pair hidden intent, parameters and noise seeds across controllers. Human forces are regenerated from the same hidden intention and the actual state; force traces are not replayed.

10.5 Optional P4-inspired reference adaptation

This is deferred, and the deferral is listed in `report.md`.

10.6 Exact vs conservative credit

In the stress test, the true friction is `μ ∈ [μ_lo, μ_hi]`. The software books `c_T = D_T(μ_lo)`, which is valid for every μ in the interval. The unreimbursed loss and the budget drain are quantified.

10.7 Post-hoc robust interval bound on the past-work error (added after the frozen run)

This bound was designed after the frozen main comparison showed the sensor-bias failure. Assume known calibration intervals `|b_f,i| ≤ B_f` and `|s − ŝ_k| ≤ B_s`. Then, per contact event, the error model of §4.2 satisfies

```
e_BH ≥ (Gaussian part) − B_f ‖Δp̂‖₁ − B_s Σ_k |G_k| h_s ,
```

The planner starts its rollouts from `BH_book − R`, with `R = B_f‖Δp̂‖₁ + B_s Σ|G_k|h_s` summed over events. The servo margin mean is shifted by `−R`. The Gaussian margin is kept, which double-counts and is therefore conservative.

The bound holds only if the intervals hold. The second-order term `ε_m·(s − ŝ)` is still dropped, so the bound is not a certificate. It is evaluated on fresh seeds and reported separately from the frozen comparison.

11. Servo-rate energy filter (implementation of §5.2 with estimated accounts)

At every 1 ms tick the software uses:

- the measured state;

- the booked ledgers;
- the belief margin of `e_BH` (mean `m_e`, std `σ_e`, refreshed at 50 Hz);
- `Ŝ` (exact model and state).

From these it forms

```
b_H = D̂_T + λ_H (BH_book + m_e − κ_H σ_e − Ŝ) ,    b_E = ĉ_T + λ_E (Ec_book − E_min) .
```

Sampled-data constraint on tick work. The naive instantaneous condition `uᵀq̇(t_k) ≤ b` was implemented first. It **leaks energy**: when the tool is nearly stalled against the human, any large force passes the test at `t_k`, then accelerates the tool and injects energy within the 1 ms hold. The result was bang–bang chattering and true BH < 0 in the first smoke test; this version is kept as the ablation `Finst`. The implemented filter instead constrains the **predicted actuator work over the tick**. It uses a second-order Taylor model of the ZOH response with the measured external generalised force `τ_e` (known contact law, measured human force at `ŝ`, known robot friction) held over the tick:

```
Δq(u) ≈ q̇ h + ½h² M⁻¹(u + τ_e),
g(u) = uᵀΔq(u)/h = uᵀq̇ + (h/2) uᵀM⁻¹u + (h/2) uᵀM⁻¹τ_e ≤ min(b_H, b_E),
u_act = argmin_{u ∈ box} ‖u − u_nom‖²_W   s.t.  g(u) ≤ min(b_H, b_E) .
```

The feasible set is convex (an ellipsoid intersected with the box) and contains `u = 0` whenever the bound is ⩾ 0. The separable KKT system gives `u_i(ν) = clip((2W_i u_nom,i − ν a_i)/(2W_i + ν c_i))`, where `a = q̇ + (h/2)M⁻¹τ_e`, `c = h·diag(M⁻¹)`. `g(u(ν))` is monotone in ν, so ν is found by bisection. Optimality is checked against SLSQP in the tests.

If even the box minimiser of `g` violates the bound, the most energy-extracting admissible input is applied and the tick is logged as `infeasible_best_effort`. C1 uses `κ_H = 0` and `m_e = 0`. Any filter modification is part of `u_act` and of booked `P_a`. The physical robot dissipation is not used in `D̂` (`D_R,lo = 0`, conservative).

This remains tier C. The Taylor remainder and the variation of the human force within a tick are not bounded, so there is no intersample proof.

12. Guarantee tiers and claim boundaries

- **(A) Proven:**
  - the contact identity `p_T = Ṡ_T + D_T` (§2);
  - the composite balance (§3);
  - the accounts inequality `V̇ ≤ −p_H` (§4);
  - the zero-input backup lemma (§5.1);
  - the continuous-time reference filter proposition under exact accounts (§5.2);
  - the exactness of the conditional Gaussian posterior under fixed gains and a conditioned trajectory (§7.3).
- **(B) Approximate:**
  - propagation through uncertain feedback motion, MC or first order (§9);
  - Gaussian tightening `mean − κ std` (§10.3);
  - the finite-feature residual;
  - the stage-Taylor intention;
  - the treatment of the Gaussian location prior on a bounded patch.

Gaussian output tightening is **not** a probabilistic certificate. A certified claim would need:

- calibrated joint input uncertainty;
- conservative nonlinear verification over a joint confidence region, or a remainder bound;
- coverage of model and calibration error;
- coverage of intersample effects.

None of these is established here.

- **(C) Empirical:** violation frequencies with Wilson/Clopper—Pearson intervals over independent episodes (never over correlated time steps), energy audits, and step–refinement convergence.

Not claimed:

- safe physical stopping, force/peak–power/injury bounds;
- recursive feasibility of the planner;
- real–time performance;
- hardware validity;
- that uncertainty or risk aversion produces compliance;
- that the four papers establish this combination.

13. Default parameters (frozen in `configs/default.json`)

group	values		
body	$m = 2.0$ kg, $I = 0.03$ kg m^2, $L \times H = 0.40 \times 0.10$ m, $r_T = (0.12, -0.10)$ m, $h_e = 0.05$ m, $s \in [-0.18, 0.18]$ m		
robot loss	$B_R = \mathrm{diag}(1.0$ N s/m, 1.0 N s/m, 0.005 N m s$)$, $c_R = 0$		
actuation		u	$\leqslant$ (40 N, 40 N, 4 N m), ZOH 1 ms
contact	$k_n = 5000$ N/m, $b_n = 150$ N s/m, $\delta_s = 0.5$ mm, $\mu = 0.3$, $v_\varepsilon = 5$ mm/s		
task	$f_d = 10$ N, $v_{nom} = 0.08$ m/s, wipe $x_{tip} \in [-0.25, 0.25]$ m, episode 9 s		
controller	$K_t = 600$ N/m, $D_t = 50$ N s/m, $k_f = 0.6$, $k_i = 8$ s^{-1}, $D_n = 40$ N s/m, $K_\theta = 40$ N m/rad, $D_\theta = 2$ N m s/rad		
human truth	$K_H = 150$ N/m, $D_H = 15$ N s/m nominal (lognormal $\sigma = 0.2$ per episode), OU $\sigma_p = 0.4$ N, $\tau_p = 0.15$ s, residual $c_g = 1$ N, $v_g = 0.05$ m/s, $F_{rel} = 35$ N		
sensors	$\sigma_m = 0.25$ N (1 kHz), location 25 Hz, $\sigma_{\varepsilon s} = 0.03$ m, $b_s \sim N(0, 0.02^2)$ m		
accounts	$E_{c0} = 0.6$ J, $B_{H0} = 0.15$ J, $E_{min} = 0$, $\lambda_E = \lambda_H = 5$ s^{-1}		
planner	$\Delta = 20$ ms, $N = 15$, substep 5 ms, $N_s = 48$, $\alpha_E = \alpha_H = 0.01$, $\beta(C4) = 2$, 18 candidates (16 + backup + brake)		
added after smoke debugging	$v_{app} = 3$ cm/s, $D_{brake} = 400$ N s/m, integrator leak 5 s^{-1}, $D_{brk} = \mathrm{diag}(40, 40, 1.5)$, $w_{res} = 2$, $R_{task} = 0.03$ J		
physics	RK4, 1 ms (refinement 0.5, 0.25 ms); servo 1 kHz in all cases		
scenarios	no_human, block_release, assist_reverse, normal_oblique (two events), moment_arm (identical normal push, s varied), ood_intent (held–out oscillating intent), sensor_bias ($b_f = (0.8, -0.5)$ N, $b_s = 6$ cm, both outside the calibrated priors), low_BH ($B_{H0} = 0.02$ J), infeasible_init (preloaded contact, $B_{H0} = 0.004$ J $< S_0$)		

GP hyperparameters are fitted offline and frozen in `outputs/models/gp_frozen.json`.